

Performance Evaluation for Deadline-Aware Hierarchical IoT Data Collection in SAGIN

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Background and Motivation

- UAV and LEO networks are important for wide-area communication and task delivery.
- Scheduling and validating communication tasks in such networks is challenging.
- Simulation-based evaluation is crucial for analyzing the performance of data collection in SAGIN.

Problem Statement

- Implement channel variation and link quality for UAV-to-UAV and UAV-to-LEO communications.
- Replay the data collection schedule in NS-3 and evaluate end-to-end communication performance.

Methods & Results

UAV-to-UAV Probe

Goal

- Intra-fleet link quality measurement
- Slave UAV → CH/master UAV

Method

- Distance → Friis path loss → SNR → Shannon rate
- UDP probing packets
- Tx/Rx bytes, delivery ratio, throughput

Output

- Per-link CSV log
- 3D position, distance, SNR
- Estimated rate, measured throughput



Link	Distance (m)	SNR (dB)	Est. Rate (Mbps)	Eff. Throughput (Mbps)
S1 → M	120	18.4	34.2	28.7
S2 → M	175	14.1	24.9	20.3
S3 → M	230	10.6	16.8	12.5

System Architecture & Data Flow

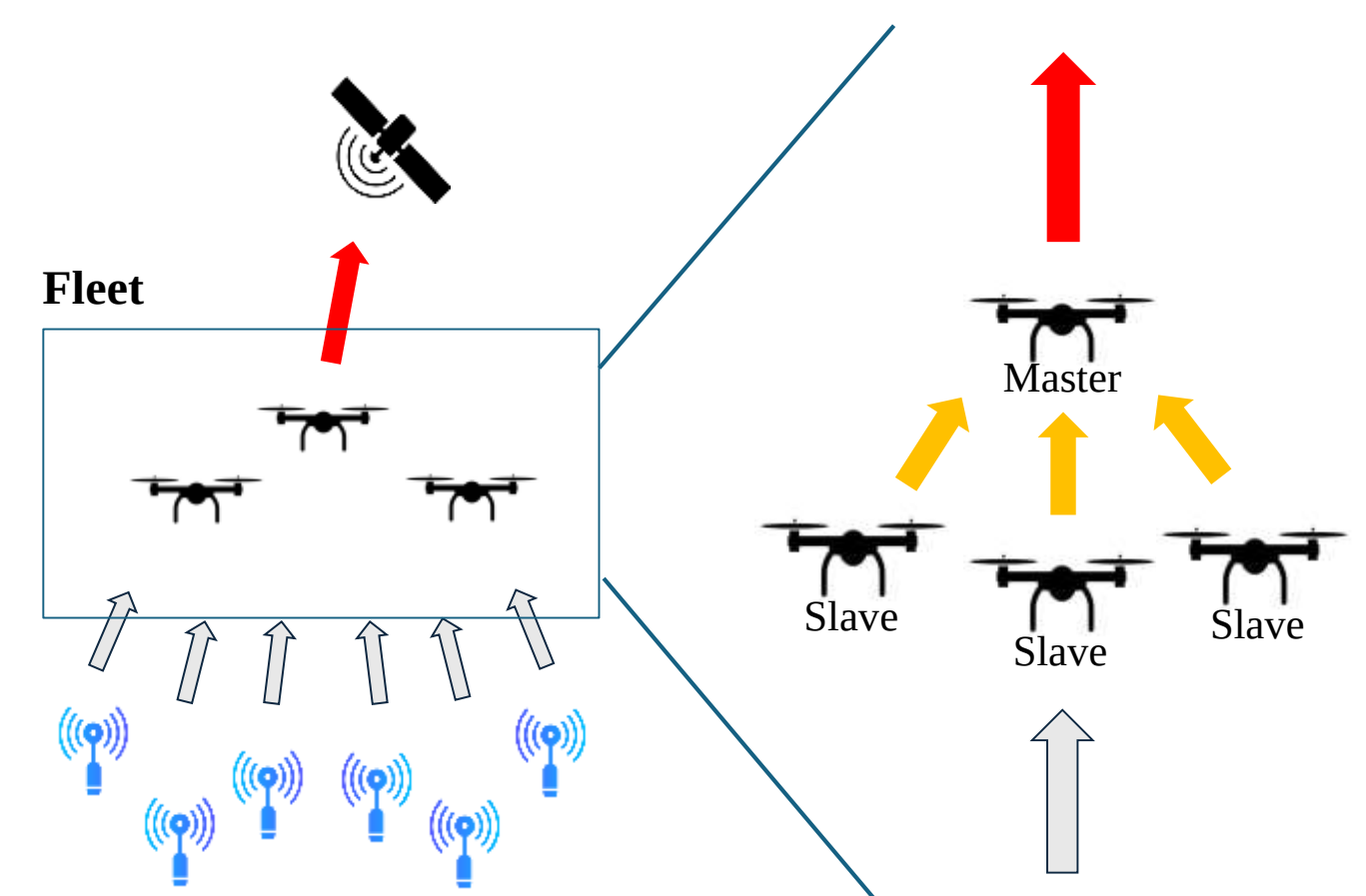


Fig. 2: System Architecture

NS-3 Schedule Replay / Verification

Goal

- Schedule feasibility validation
- Trace-driven NS-3 replay

Input

- Scheduler JSON
- MOVE / ROLE_SET
- SENSOR_TO_UAV/SLAVE_TO_MASTER

Verification

- UAV mobility and role changes
- UAV buffer dependency
- Task completion and deadline check

Output

- Completed tasks
- Mission finish time
- Deadline satisfaction

Scenario	UAVs	Sensors	Transfers	Completed	Deadline Met
Simple trace	3	50	79	79 / 79	Yes
Normal trace	5	500	212	212 / 212	Yes

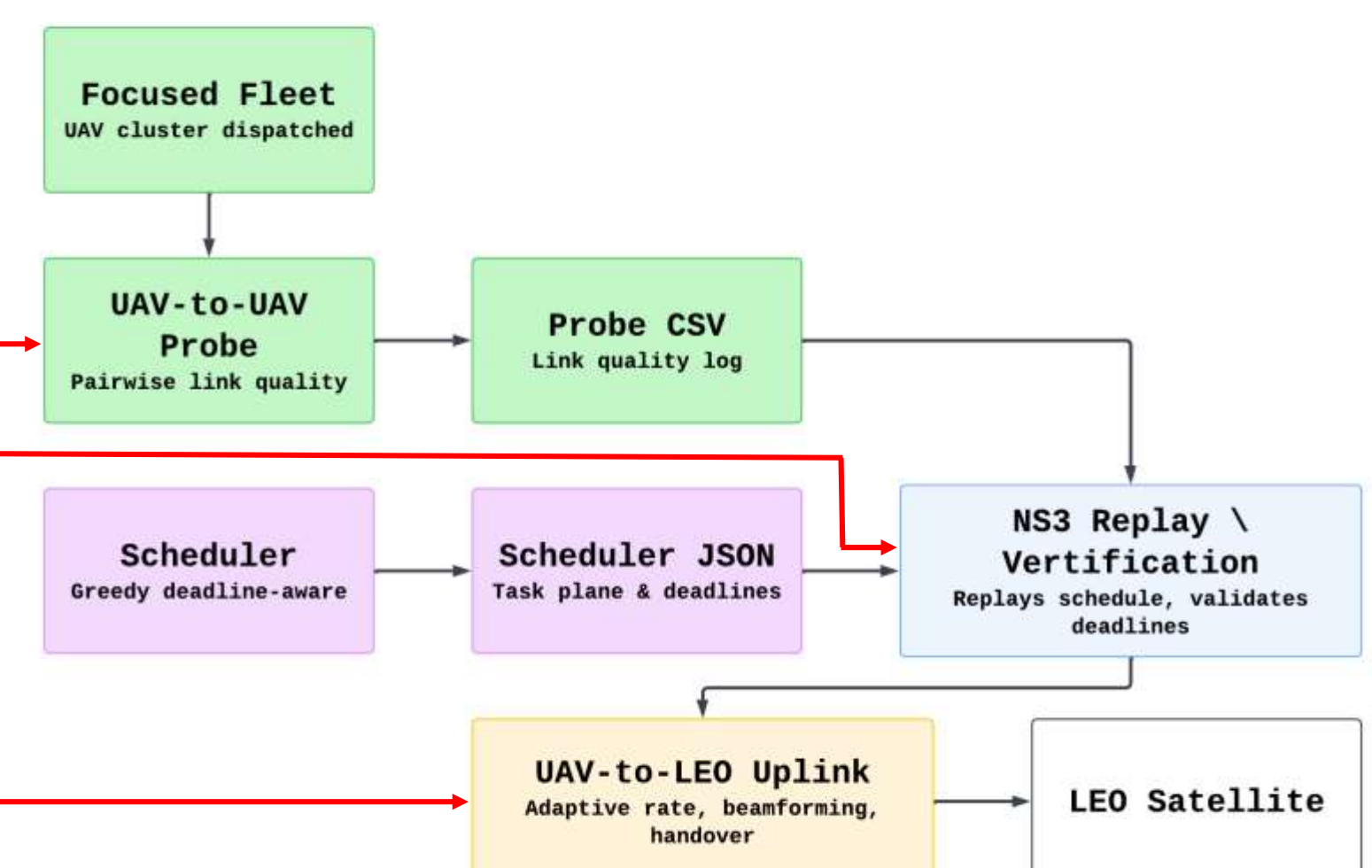


Fig. 3: Data Flow

UAV-to-LEO Uplink

Goal

Establish high-capacity uplink between Cluster Head UAV and fast-moving LEO satellite

Method

1. Link Budget Model: Pipeline: FSPL → SNR → Shannon capacity.
2. Hybrid Beamforming Gain (Fig. 1)
3. SNR-Adaptive Rate Control

Output

- Link related data (e.g. SNR)
- Visible time window

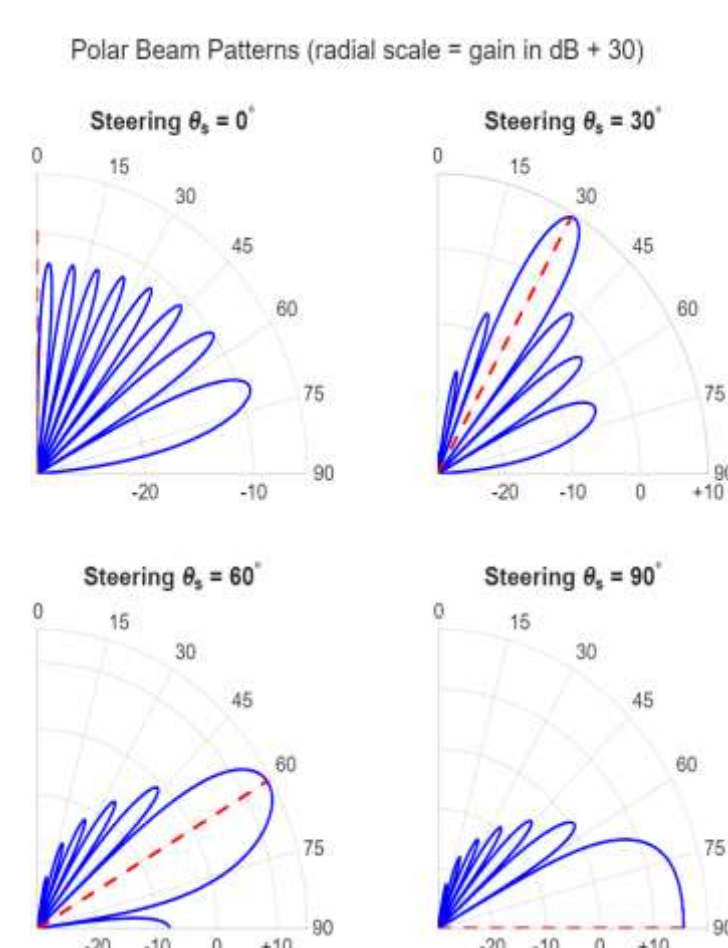


Fig. 1: Polar Beamforming Patterns

Conclusion

- We built an NS-3-based framework to support link probing, schedule replay, and end-to-end performance measurement in SAGIN scenarios.
- The framework provides a platform for analyzing various scheduling decisions.
- Future work: integrate scheduling algorithm into the simulation framework to enable interactive network planning